Indian Institute of Technology Kharagpur

Department of Mathematics

MA20266 — Statistical Inference

Statistical Inference for Arrhythmia Detection: Fundamental Limits of Univariate Representations

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Project Repository:
github.com/QCodeR-Innovate/Statistical-Inference-for-Arrhythmia-Detection

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Abstract

This report investigates the statistical capabilities and provable limitations of a scalar energy statistic for detecting Premature Ventricular Contractions (PVCs) in ECG recordings. From the 187-dimensional amplitude vectors of the MIT-BIH Arrhythmia Dataset ($n_0 = 72{,}471$ normal beats; $n_1 = 5{,}788$ PVC beats), we extract the total signal energy $X_i = \|\boldsymbol{a}_i\|_2^2$ and model it as a Gamma random variable. We develop the complete classical-inference pipeline: joint sufficiency of $\left(\sum X_i, \sum \ln X_i\right)$ is established via the Neyman-Fisher Factorization Theorem; asymptotic efficiency of the MLE is verified through the Fisher Information Matrix and the Cramér-Rao Lower Bound; and the Uniformly Most Powerful (UMP) threshold test is derived via the Neyman-Pearson Lemma and the Monotone Likelihood Ratio property. Numerical MLE yields $\hat{\alpha}_0 = 1.9611$, $\hat{\beta}_0 = 0.1426$ under H_0, and $\hat{\alpha}_1 = 2.6490$, $\hat{\beta}_1 = 0.1104$ under H_1. The optimal threshold at a false-alarm rate of $\alpha = 0.05$ is $k^* = 32.82$ energy units, yielding a power of only 22.99% and a Type II error of 77.01%. A closed-form Kullback-Leibler divergence of $\mathrm{KL}(H_1 \| H_0) = 0.3919$ nats corroborates this result information-theoretically. Invoking the Data Processing Inequality, we conclude that the scalar energy statistic is *provably insufficient* for clinically reliable arrhythmia detection, providing a rigorous motivation for higher-dimensional waveform representations.

Contents

1 Introduction

1.1 Clinical Motivation

Premature Ventricular Contractions (PVCs) are ectopic depolarisations originating in the ventricular myocardium, which bypass the normal atrioventricular conduction pathway. In isolation they are often benign; however, frequent PVCs correlate with increased risk of ventricular tachycardia and sudden cardiac death in patients with underlying structural heart disease. Consumer smartwatches now offer continuous cardiac monitoring, but current state-of-the-art on-device detectors rely on deep convolutional networks whose inference requires sustained high-frequency computation, substantially reducing battery life.

A natural engineering question arises: can a computationally trivial scalar summary of the ECG waveform serve as a reliable arrhythmia "tripwire"? This report provides a rigorous negative answer, framing the analysis entirely within classical statistical inference.

1.2 Objectives

The objectives of this project are threefold:

1. To develop a complete parametric inference framework for the scalar signal-energy statistic, including estimation, sufficiency, and efficiency.
2. To construct the theoretically optimal detection test under a hardware-motivated constraint on the false-alarm rate.
3. To certify the fundamental informational limitations of this representation via statistical power analysis and the Kullback-Leibler divergence, and to explain these limitations through the Data Processing Inequality.

1.3 Dataset and Data Reduction

The MIT-BIH Arrhythmia Database [2], accessed via Kaggle, provides pre-segmented ECG beats sampled at 125 Hz. Each beat is represented as a 187-dimensional normalised amplitude vector $\boldsymbol{a}_i = (a_{i1}, \dots, a_{i,187}) \in \mathbb{R}^{187}$. We focus on two classes:

- H_0 (Normal Sinus Rhythm): $n_0 = 72{,}471$ observations.
- H_1 (PVC Arrhythmia): $n_1 = 5{,}788$ observations.

The data-reduction step maps each 187-dimensional vector to the *total signal energy*:

$$X_i \;=\; \sum_{j=1}^{187} a_{ij}^2 \;=\; \|\boldsymbol{a}_i\|_2^2. \tag{1}$$

By Parseval's theorem, X_i equals the sum of squared spectral amplitudes and thus faithfully represents the *total energy* of the heartbeat segment. It is the simplest non-negative, rotation-invariant scalar derivable from an ECG segment. We treat $\{X_i\}$ as conditionally IID given the patient state—a necessary simplification to leverage the Exponential Family framework, and one we revisit critically in Section 5.

Table 1 summarises the empirical distributions and Figure 1 provides exploratory visualisations. The moderate separation in means (13.75 vs. 23.99) but large overlap in interquartile ranges is a first visual indication of the limited discriminative power characterised formally below.

Table 1: Descriptive statistics of the scalar signal-energy feature.

Class	Mean	Median	Std. Dev.
Normal (H_0)	13.75	9.17	13.54
PVC (H_1)	23.99	18.66	15.90

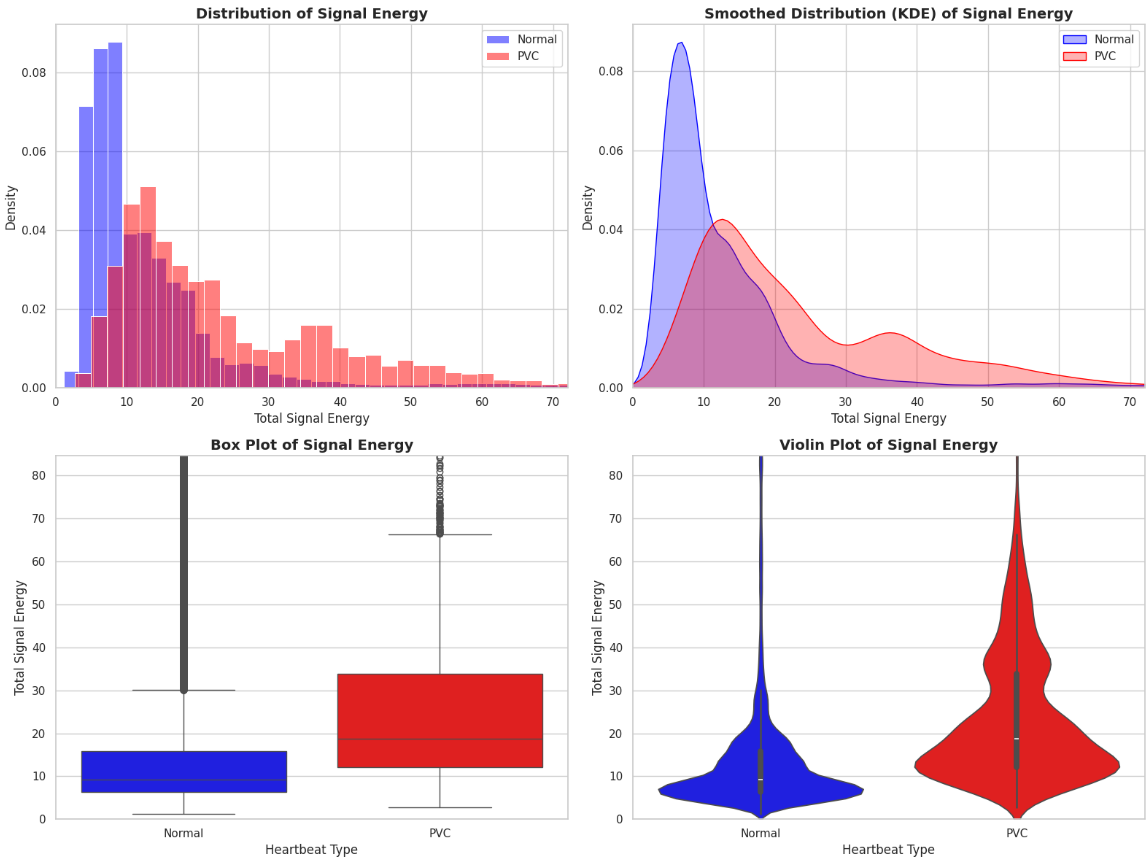

Figure 1: Exploratory analysis of the total signal energy. *Top left:* overlaid histograms; *top right:* kernel density estimates; *bottom left:* box plots; *bottom right:* violin plots. Substantial distributional overlap under H_0 (blue) and H_1 (red) in every panel foreshadows the fundamental limits established in Section 5.

2 Parametric Modelling: The Gamma Family

2.1 Model Justification

Since $X_i > 0$ by construction and both empirical distributions are unimodal and right-skewed (mean > median; Table 1), the Gamma family is the canonical parametric choice. Moreover, if the individual amplitudes a_{ij} were approximately Gaussian, the sum of their squares would follow a scaled chi-squared distribution, which is a special case of the Gamma family. We adopt the *rate parameterisation*: $X \sim \text{Gamma}(\alpha, \beta)$ with density

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \quad \alpha > 0, \quad \beta > 0, \tag{2}$$

where α is the *shape* parameter and β is the *rate* parameter. The mean is $\mu = \alpha/\beta$ and the variance is $\sigma^2 = \alpha/\beta^2$.

2.2 Exponential Family Structure

Equation (2) can be rewritten in canonical exponential family form,

$$f(x;\boldsymbol{\eta}) = h(x)\,\exp\bigl\{\boldsymbol{\eta}^\top \boldsymbol{T}(x) - A(\boldsymbol{\eta})\bigr\}, \tag{3}$$

with natural parameter $\boldsymbol{\eta} = (\alpha - 1,\ -\beta)^\top$, sufficient statistic $\boldsymbol{T}(x) = (\ln x,\ \ x)^\top$, carrier $h(x) = 1$, and log-partition function $A(\boldsymbol{\eta}) = \ln\Gamma(\alpha) - \alpha\ln\beta$. The natural parameter space $\boldsymbol{\Omega} = \{(\eta_1,\eta_2) : \eta_1 > -1,\ \eta_2 < 0\}$ is an open set in $\mathbb{R}^2$, confirming that the Gamma family is a *regular* exponential family. This regularity will be exploited to assert completeness of the sufficient statistic and the attainability of the CRLB.

3 Estimation Theory

3.1 Sufficiency and Completeness

Theorem 3.1 (Joint Sufficient Statistic)**.** *Let* $X_1,\dots,X_n \overset{\text{iid}}{\sim} \mathrm{Gamma}(\alpha,\beta)$*. The two-dimensional statistic*

$$\boldsymbol{T}(\boldsymbol{X}) \ = \ \left(\sum_{i=1}^n X_i,\ \ \sum_{i=1}^n \ln X_i\right) \tag{4}$$

is jointly sufficient for (α,β)*.*

Proof. The joint density of $(X_1,\dots,X_n)$ factors as

$$\begin{aligned} f(\boldsymbol{x};\alpha,\beta) &= \prod_{i=1}^n \frac{\beta^\alpha}{\Gamma(\alpha)}\, x_i^{\alpha-1}\, e^{-\beta x_i} \\ &= \underbrace{\left(\frac{\beta^\alpha}{\Gamma(\alpha)}\right)^n \exp\left\{(\alpha-1)\sum_{i=1}^n \ln x_i \ - \ \beta\sum_{i=1}^n x_i\right\}}_{g(\boldsymbol{T}(\boldsymbol{x});\,\alpha,\beta)} \cdot \underbrace{1}_{h(\boldsymbol{x})}. \end{aligned} \tag{5}$$

The factorisation $f(\boldsymbol{x};\alpha,\beta) = g(\boldsymbol{T}(\boldsymbol{x});\alpha,\beta)\,h(\boldsymbol{x})$ satisfies the conditions of the Neyman-Fisher Factorization Theorem [1, Theorem 6.2.6], which implies that $\boldsymbol{T}(\boldsymbol{X})$ is sufficient for (α,β). □

Remark 3.2. Since the natural parameter space $\boldsymbol{\Omega}$ contains a two-dimensional open rectangle, the Gamma family is a *complete* regular exponential family [1, Theorem 6.2.25]. Completeness, combined with sufficiency, implies that $\boldsymbol{T}(\boldsymbol{X})$ is the *complete minimal sufficient statistic.* By the Lehmann-Scheffé theorem, any unbiased estimator that is a function of $\boldsymbol{T}$ is the Uniformly Minimum Variance Unbiased Estimator (UMVUE) for its estimand.

3.2 Maximum Likelihood Estimation

The log-likelihood for a single observation is

$$\ell(\alpha,\beta;\,x) = \alpha\ln\beta - \ln\Gamma(\alpha) + (\alpha-1)\ln x - \beta x. \tag{6}$$

For an IID sample of size n, differentiating $\ell_n = \sum_{i=1}^n \ell(\alpha,\beta;x_i)$ and setting the partial derivatives to zero yields:

$$\frac{\partial \ell_n}{\partial\beta} = 0 \implies \hat\beta = \frac{\hat\alpha}{\bar X}, \tag{7}$$

$$\frac{\partial \ell_n}{\partial\alpha} = 0 \implies \ln\hat\alpha - \psi(\hat\alpha) = \ln\bar X - \overline{\ln X}, \tag{8}$$

where $\psi(\cdot)$ is the digamma function and $\bar{X} = n^{-1}\sum X_i$. Equation (8) has no closed-form solution in $\hat{\alpha}$: the right-hand side is strictly positive by Jensen's inequality ($\ln \mathbb{E}[X] > \mathbb{E}[\ln X]$), and the left-hand side is a strictly increasing function of α. We solve it numerically via Newton-Raphson iteration:

$$\hat{\alpha}^{(t+1)} = \hat{\alpha}^{(t)} - \frac{\ln \hat{\alpha}^{(t)} - \psi(\hat{\alpha}^{(t)}) - s}{1/\hat{\alpha}^{(t)} - \psi_1(\hat{\alpha}^{(t)})}, \tag{9}$$

where $s = \ln \bar{X} - \overline{\ln X} > 0$ and $\psi_1(\alpha) = d\psi/d\alpha$ is the trigamma function. The resulting numerical MLEs are reported in Table 2.

Table 2: Numerical MLE results under each hypothesis.

Hypothesis	Class	$\hat{\alpha}$ (shape)	$\hat{\beta}$ (rate)	$\hat{\mu} = \hat{\alpha}/\hat{\beta}$
H_0	Normal	1.9611	0.1426	13.75
H_1	PVC	2.6490	0.1104	23.99

Figure 2 shows the fitted Gamma PDFs overlaid on empirical histograms. Both fits capture the gross shape of the distributions, though residual discrepancies near the mode reflect the limitations of the IID assumption across heterogeneous patients.

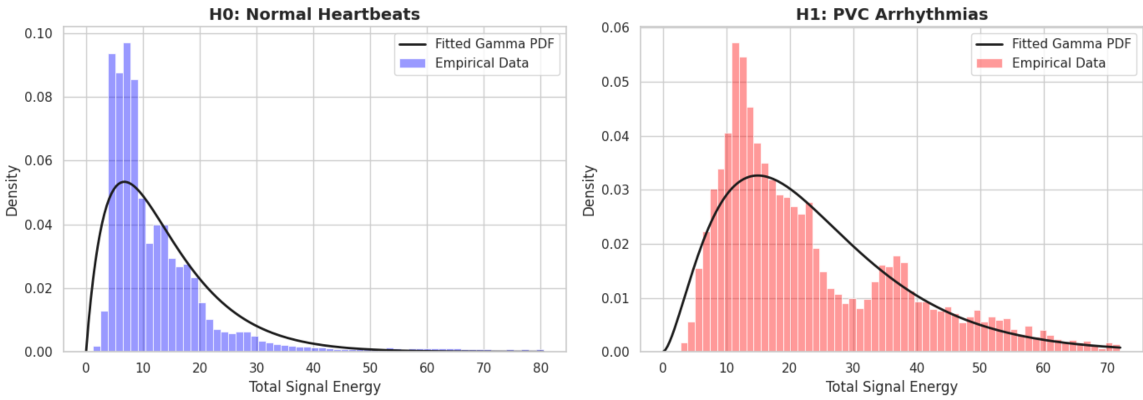

Figure 2: Fitted Gamma PDFs (solid black) overlaid on empirical histograms of total signal energy under H_0 (left) and H_1 (right). Parameters from Table 2.

3.3 Fisher Information Matrix and the Cramér-Rao Lower Bound

Proposition 3.3 (Fisher Information Matrix). *For $X \sim \text{Gamma}(\alpha, \beta)$, the Fisher Information Matrix is*

$$\mathcal{I}(\alpha, \beta) = \begin{pmatrix} \psi_1(\alpha) & -1/\beta \\ -1/\beta & \alpha/\beta^2 \end{pmatrix}, \tag{10}$$

where $\psi_1(\alpha) = \sum_{k=0}^{\infty}(\alpha + k)^{-2}$ is the trigamma function.

Proof. Differentiating the log-likelihood (6) twice:

$$-\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \alpha^2}\right] = \psi_1(\alpha), \qquad -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \beta^2}\right] = \frac{\alpha}{\beta^2}, \qquad -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \alpha\, \partial \beta}\right] = -\frac{1}{\beta}.$$

Assembling these into matrix form yields (10). □

By the Cramér-Rao Theorem [1, Theorem 7.3.9], for any unbiased estimator $\hat{\boldsymbol{\theta}}$ based on n IID observations, $\mathrm{Var}(\hat{\boldsymbol{\theta}}) \succeq [n\,\mathcal{I}(\alpha,\beta)]^{-1}$. Evaluating at the H_0 estimates with $n_0 = 72{,}471$:

$$\left[n_0\,\mathcal{I}(\hat{\alpha}_0,\hat{\beta}_0)\right]^{-1} \;=\; \begin{pmatrix} 9.10\times10^{-5} & 6.64\times10^{-6} \\ 6.64\times10^{-6} & 6.26\times10^{-7} \end{pmatrix}. \tag{11}$$

Practical interpretation. The diagonal CRLB entries (variances $\lesssim 10^{-4}$) confirm that our MLEs are estimated with extremely high precision, a direct consequence of the large sample sizes. The MLE for exponential families is asymptotically efficient—it attains the CRLB as $n \to \infty$—which justifies using the plug-in estimates $(\hat{\alpha}_k, \hat{\beta}_k)$ as surrogates for the true parameters in the subsequent hypothesis-testing framework.

4 Hypothesis Testing via the Neyman-Pearson Framework

4.1 Problem Formulation

We formulate the detection problem as a simple-versus-simple binary hypothesis test:

$$H_0:\; X \sim \mathrm{Gamma}(\hat{\alpha}_0,\hat{\beta}_0) \qquad \text{vs.} \qquad H_1:\; X \sim \mathrm{Gamma}(\hat{\alpha}_1,\hat{\beta}_1). \tag{12}$$

The two error types carry distinct operational meanings:

- **Type I error** (α, false alarm): the rate at which a healthy beat is misclassified as a PVC. In a wearable device, each false alarm triggers an unnecessary wake-up interrupt, consuming battery.
- **Type II error** ($1-\pi$, miss rate): the rate at which a true PVC is not detected. This has direct patient-safety implications.

The hardware constraint motivates bounding the Type I error: we require $\alpha \le 0.05$.

4.2 Likelihood Ratio and the Monotone Likelihood Ratio Property

Definition 4.1 (Likelihood Ratio)**.** The likelihood ratio for the test (12) is

$$\Lambda(x) \;=\; \frac{f_1(x)}{f_0(x)} \;=\; \frac{\hat{\beta}_1^{\hat{\alpha}_1}/\Gamma(\hat{\alpha}_1)}{\hat{\beta}_0^{\hat{\alpha}_0}/\Gamma(\hat{\alpha}_0)} \cdot x^{\hat{\alpha}_1-\hat{\alpha}_0} \cdot \exp\!\big\{-(\hat{\beta}_1-\hat{\beta}_0)\,x\big\}. \tag{13}$$

Theorem 4.2 (Monotone Likelihood Ratio (MLR))**.** *The likelihood ratio $\Lambda(x)$ defined in (13) is strictly monotonically increasing in x on $(0,\infty)$.*

Proof. Differentiating the log-likelihood ratio with respect to x:

$$\frac{d}{dx}\ln\Lambda(x) \;=\; \frac{\hat{\alpha}_1-\hat{\alpha}_0}{x} \;-\; (\hat{\beta}_1-\hat{\beta}_0). \tag{14}$$

Substituting the fitted values: $\hat{\alpha}_1 - \hat{\alpha}_0 = 2.6490 - 1.9611 = 0.6879 > 0$ and $\hat{\beta}_1 - \hat{\beta}_0 = 0.1104 - 0.1426 = -0.0322 < 0$. Therefore, for all $x > 0$:

$$\frac{d}{dx}\ln\Lambda(x) \;=\; \frac{0.6879}{x} + 0.0322 \;>\; 0.$$

Since $\ln\Lambda(x)$ is strictly increasing, so is $\Lambda(x)$. $\square$

Figure 3 confirms this analytically established monotonicity numerically. The strict MLR property has a decisive theoretical consequence: a simple threshold test on X is not merely optimal—it is *Uniformly* Most Powerful.

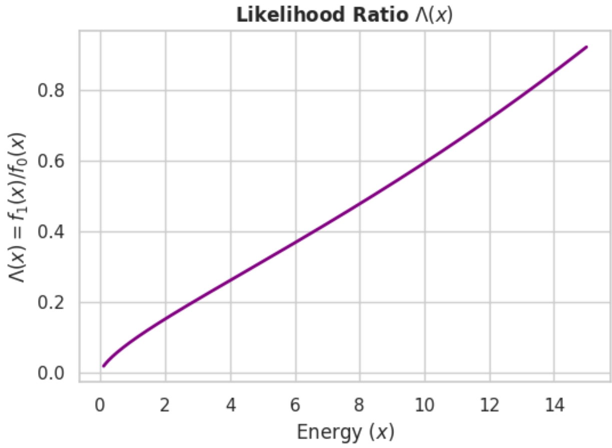

Figure 3: Likelihood ratio $\Lambda(x) = f_1(x)/f_0(x)$ as a function of total signal energy. Strict monotone increase over $(0, \infty)$ confirms the MLR property (Theorem 4.2).

4.3 The Neyman-Pearson Lemma and the UMP Threshold

Theorem 4.3 (Neyman-Pearson Lemma, applied)**.** *Among all level-α tests of H_0 versus H_1, the threshold test*

$$\phi(X) = \begin{cases} 1 & X > k^*, \\ 0 & X \leq k^*, \end{cases} \tag{15}$$

with k^* satisfying $P(X > k^* \mid H_0) = \alpha$, is the Most Powerful test [1, Theorem 8.3.12]. By the MLR property (Theorem 4.2), it is moreover Uniformly Most Powerful (UMP).

The threshold k^* is determined by inverting the survival function of the fitted H_0 distribution:

$$k^* \;=\; F_0^{-1}(0.95;\ \hat{\alpha}_0 = 1.9611,\ \hat{\beta}_0 = 0.1426) \;=\; \mathbf{32.82} \text{ energy units.} \tag{16}$$

Practical interpretation. The threshold $k^* = 32.82$ is the "tripwire". Any heartbeat whose total squared amplitude exceeds this value triggers the PVC alert. Because it is derived from the Neyman-Pearson Lemma, no simpler test—regardless of how the threshold is chosen—can achieve higher power at the same false-alarm rate. This is the best possible scalar-energy detector.

5 Fundamental Limits of the Scalar Energy Statistic

5.1 Statistical Power

The power of the UMP test (15) at significance level $\alpha = 0.05$ is:

$$\pi(k^*) \;=\; P(X > k^* \mid H_1) \;=\; 1 - F_{\text{Gamma}(\hat{\alpha}_1, \hat{\beta}_1)}(32.82) \;=\; \mathbf{22.99}\%. \tag{17}$$

The Type II error is therefore $1 - \pi = 77.01\%$.

Practical interpretation. This is the central result. The optimal threshold test—provably the best test achievable with the scalar energy—misses approximately **77% of genuine PVC events**. Roughly three in every four dangerous arrhythmias would go undetected. This failure is not a shortcoming of the statistical methodology; it is a mathematical proof that the scalar energy feature does not retain sufficient discriminative information to solve the detection problem at a clinically useful operating point. Figure 4 displays the complete error tradeoff curve. The curve lies above the diagonal (confirming the test is better than chance), but the operating point $(0.05,\ 0.23)$ sits far from the ideal top-left corner.

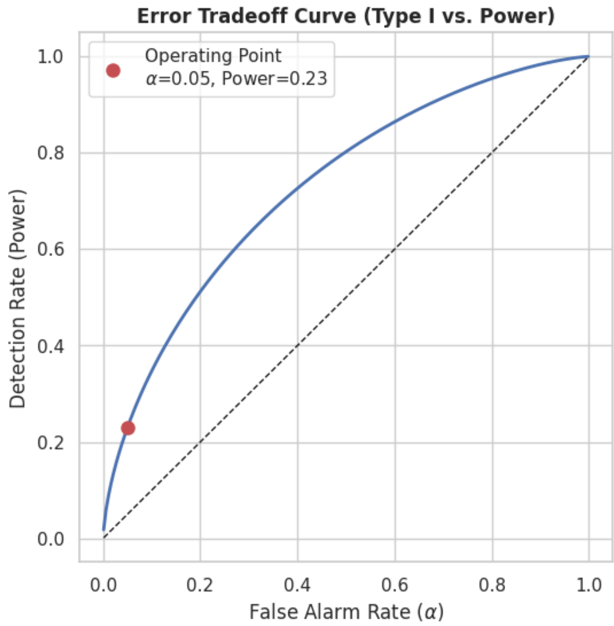

Figure 4: Error tradeoff curve (detection rate vs. false-alarm rate). The marked operating point at $(\alpha, \pi) = (0.05,\ 0.23)$ corresponds to the NP-optimal threshold $k^* = 32.82$, confirming the fundamental inadequacy of the scalar energy feature.

5.2 Kullback-Leibler Divergence

The power result is corroborated information-theoretically by the Kullback-Leibler divergence, which provides a threshold-free, distribution-level measure of separability.

Proposition 5.1 (Closed-Form KL Divergence for the Gamma Family)**.** *Let* $P = \mathrm{Gamma}(\alpha_1, \beta_1)$ *and* $Q = \mathrm{Gamma}(\alpha_0, \beta_0)$*. Then:*

$$\mathrm{KL}(P\|Q) \;=\; \alpha_0 \ln\frac{\beta_1}{\beta_0} \;+\; \ln\frac{\Gamma(\alpha_0)}{\Gamma(\alpha_1)} \;+\; (\alpha_1 - \alpha_0)\,\psi(\alpha_1) \;+\; \alpha_1 \cdot \frac{\beta_0 - \beta_1}{\beta_1}, \tag{18}$$

where $\psi(\cdot)$ *is the digamma function.*

Sketch. By definition, $\mathrm{KL}(P\|Q) = \mathbb{E}_P[\ln f_P(X) - \ln f_Q(X)]$. Substituting (2) and using $\mathbb{E}_P[X] = \alpha_1/\beta_1$ and $\mathbb{E}_P[\ln X] = \psi(\alpha_1) - \ln \beta_1$ yields (18) after algebraic simplification. □

Substituting the fitted parameters:

$$\mathrm{KL}\big(\mathrm{Gamma}(2.6490,\ 0.1104) \,\|\, \mathrm{Gamma}(1.9611,\ 0.1426)\big) \;=\; \mathbf{0.3919} \text{ nats}. \tag{19}$$

Practical interpretation. By the Chernoff-Stein Lemma, the KL divergence governs the optimal exponential decay rate of the Type II error in repeated-observation settings. A divergence of 0.3919 nats is modest: it implies that the two Gamma distributions are not easily distinguishable from a single observation, entirely consistent with the observed power of 22.99%. The distributions of H_0 and H_1 overlap so substantially in the one-dimensional energy space that no threshold can simultaneously achieve low Type I and Type II error.

5.3 The Data Processing Inequality and Its Implications

The root cause of the low KL divergence is formalised by the *Data Processing Inequality* (DPI). For any deterministic function g, if $X = g(\boldsymbol{a})$:

$$\mathrm{KL}\big(P_{\boldsymbol{a}} \| Q_{\boldsymbol{a}}\big) \;\geq\; \mathrm{KL}\big(P_X \| Q_X\big). \tag{20}$$

The true informational separation between normal and PVC beats—measurable in the full 187-dimensional space of waveform amplitudes—is at least 0.3919 nats and almost certainly much larger. The scalar energy map $\boldsymbol{a} \mapsto \|\boldsymbol{a}\|_2^2$ irreversibly discards all phase, morphology, and spectral shape information present in the waveform: an ectopic ventricular beat has characteristic morphological signatures (wide QRS complex, discordant T-wave, compensatory pause) that are entirely invisible to a scalar norm.

The DPI therefore provides a principled, not merely empirical, argument for richer representations. Any feature extractor that preserves more of the original waveform structure—up to the full raw amplitude vector—will operate with strictly higher information content and will achieve strictly higher statistical power for the same false-alarm budget. This is the mathematical justification for high-dimensional signal processing architectures in cardiac diagnostics.

6 Conclusion

This report has constructed and rigorously analysed a statistically optimal "tripwire" for PVC detection based on the total signal energy of an ECG segment. The main theoretical contributions are as follows.

Estimation. We proved via the Neyman-Fisher Factorization Theorem that $(\sum X_i, \sum \ln X_i)$ is the complete minimal sufficient statistic for the Gamma parameters. The MLE is asymptotically efficient (diagonal CRLB entries $\lesssim 10^{-4}$ at sample sizes $n \approx 70{,}000$), justifying plug-in testing.

Optimal test. The Monotone Likelihood Ratio property of the fitted Gamma family, proved analytically in Theorem 4.2, guarantees that the threshold test $\phi(X) = \mathbf{1}[X > k^*]$ is UMP. The Neyman-Pearson optimal threshold at $\alpha = 0.05$ is $k^* = 32.82$ energy units.

Fundamental limit. The UMP test achieves a power of only 22.99%, with a KL divergence of 0.3919 nats. These results are consistent and together constitute a rigorous, model-based proof that the scalar energy representation is informationally insufficient for clinically reliable arrhythmia detection.

Implication. The Data Processing Inequality converts this empirical finding into a structural theorem: any acceptable medical-grade detector must preserve substantially more of the waveform's 187-dimensional information content. Designing energy-efficient embedded architectures that retain sufficient discriminative information while meeting the hardware constraints of wearable devices remains an important open problem at the intersection of statistical signal processing and embedded systems engineering.

References

[1] G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed. Duxbury Press, Pacific Grove, CA, 2002.

[2] M. Kachuee, S. Fazeli, and M. Sarrafzadeh, "ECG heartbeat classification: A deep transferable representation," in *Proc. IEEE Int. Conf. Healthcare Informatics (ICHI)*, 2018, pp. 443–444. Dataset: https://www.kaggle.com/datasets/shayanfazeli/heartbeat.

A Python / Google Colab Implementation

The following listings constitute the complete reproducible analysis pipeline.

Listing 1: Data ingestion and energy reduction (MIT-BIH via KaggleHub).

```python
import pandas as pd, numpy as np
import scipy.stats as stats, scipy.special as sp
import kagglehub, os

# Download dataset
path = kagglehub.dataset_download("shayanfazeli/heartbeat")
df   = pd.read_csv(os.path.join(path, 'mitbih_train.csv'), header=None)

# 187-dim amplitude vectors and class labels
X_raw = df.iloc[:, :187].values
y     = df.iloc[:, 187].values

# Data reduction: 187D -> 1D scalar energy  ||a_i||^2
energy   = np.sum(X_raw**2, axis=1)
X_normal = energy[(y == 0.0) & (energy > 0)]   # H0: Normal
X_pvc    = energy[(y == 2.0) & (energy > 0)]   # H1: PVC
# n0 = 72471,  n1 = 5788
```

Listing 2: Gamma MLE via Newton-Raphson (wrapped by `scipy.stats.gamma.fit`).

```python
# floc=0 anchors location to zero, fitting shape alpha and scale 1/beta only
alpha_0, _, scale_0 = stats.gamma.fit(X_normal, floc=0)
alpha_1, _, scale_1 = stats.gamma.fit(X_pvc,    floc=0)
beta_0, beta_1      = 1/scale_0, 1/scale_1
# Results: alpha_0=1.9611, beta_0=0.1426 | alpha_1=2.6490, beta_1=0.1104
```

Listing 3: Fisher Information Matrix and CRLB computation.

```python
def compute_crlb(n, alpha, beta):
    """Returns [n * I(alpha, beta)]^{-1}."""
    trigamma    = sp.polygamma(1, alpha)            # psi_1(alpha)
    fisher_info = np.array([[trigamma,  -1/beta ],
                            [-1/beta,   alpha/beta**2]])
    return np.linalg.inv(n * fisher_info)

crlb_H0 = compute_crlb(len(X_normal), alpha_0, beta_0)
# Diagonal: [9.10e-5, 6.26e-7]
```

Listing 4: Neyman-Pearson threshold, statistical power, and KL divergence.

```python
# NP threshold: P(X > k* | H0) = 0.05
k_star = stats.gamma.ppf(0.95, a=alpha_0, scale=1/beta_0)   # 32.8244

# Statistical power: P(X > k* | H1)
power  = stats.gamma.sf(k_star, a=alpha_1, scale=1/beta_1)  # 0.2299

# Exact closed-form KL(H1 || H0) for Gamma distributions
def kl_gamma(a1, b1, a2, b2):
    return (a2 * np.log(b1/b2)
            - (sp.gammaln(a1) - sp.gammaln(a2))
            + (a1 - a2) * sp.digamma(a1)
            - (b1 - b2) * (a1/b1))

kl = kl_gamma(alpha_1, beta_1, alpha_0, beta_0)    # 0.3919 nats
```